

ECE 585
Term Project Instructions
March 12, 2026

Professor E. Rosenbaum

Spring Semester 2026

Overview

Working singly or—preferably—in groups of two, students will carry out an original research project or write a survey paper on a topic that is relevant to the class material. At the end of the term, each student will give an oral presentation on his/her project and submit a final report.

Following Problem Set #6, the homework assignments will be shorter, so that you have additional time available to work on the project.

Deadlines

9:30 AM, Thursday, March 26: One-half page description of your proposed project.

(Submit as an email attachment to Spring_.7zk6gzdku6lx3ens@u.box.com)

The research topic, approach and expected results must be described. The instructor has the option of vetoing a project topic if she deems the likelihood of success to be low, or if the topic is not relevant to the class material, or if it is too similar to a project proposed by another student. Because the project topic is subject to instructor approval, students are cautioned against doing substantive work before handing in this assignment and obtaining the instructor's approval. **Students are welcome (encouraged) to submit their proposals prior to the deadline, and proposals will be reviewed as they are submitted.**

Thursday, April 30: First set of in-class oral presentations. (2 additional sessions will follow. See course information sheet for details.)

15 minutes will be allocated to each project: 10-12 minute powerpoint presentation and 3-5 minutes for questions. A laptop computer and projector will be available; you must email your powerpoint slides to Prof. Rosenbaum **no later** than 8:00 AM on the day of your presentation.

11:59 PM, Friday, May 8: Deadline to submit written project report. (Upload to Gradescope.)

Instructions

The report must be typed (word processed) using an IEEE journal paper template; MS Word and LaTeX versions of the templates should be downloaded from the class Canvas page. The content of the report and its page length limitations are described below.

Abstract: This appears following the paper title and author name(s). See IEEE publications such as *Electron Device Letters* for examples of abstracts.

Text: **Four** pages **maximum** for original research projects. A list of references is expected and will not be counted towards the four-page limit. For survey papers, the page limit is **eight**; as in the previous case, there is no penalty if the references spill beyond the 8th page.

Figures: The generous use of figures is highly recommended. Each should have a descriptive caption below it. There is no limit on the allowable number of figures, but you must select the most important figures for inclusion in the main body of your paper, which is subject to a page length limitation. Additional figures and tables may be placed in an appendix.

Appendices: Any number of pages may be attached as appendices. This is where you may show lengthy derivations of equations. It is not necessary to include any appendices.

Grading

70% for technical content and 30% for presentation. For original research projects, technical content includes correctness of physics and mathematics, significance of problem, innovation in approach, and usefulness of results. Survey papers will be judged to have good technical content if the student has done a

thorough, critical, and rigorous evaluation of previous works; also, if a technologically significant topic has been investigated. A survey paper may be judged unfavorably if it has a too small bibliography or one that does not include recent works. Written reports are due roughly one week after the oral presentations; students are expected to use the technical feedback provided by the audience to improve their final, written reports. Presentation quality has the following components: grammar, organization, clarity, effective use of examples and figures. Both the oral and written presentations will be evaluated. Your fellow students will rate your oral presentation, and their evaluations will be used in assigning the grade. Note that it is difficult to impress people with your presentation if the technical content is poor.

Collaboration

Two students may conduct an original research project and submit a joint term paper. Grading will be based on the quality of the work and is unaffected by whether one person or two persons did the work. If two students wish to collaborate on a survey paper, a compelling rationale must be provided. For example, the pair of students might explain that the scope of the survey is very broad, or the students might propose to compare and contrast two different models, with each student being responsible for developing deep familiarity with one of the two models.

Project Content

The subject material must be related to the course material. In most cases, you will need to start by doing a literature review to become familiar with existing work.

By the time that the project proposals are due, we should have completed our discussions of DC and transient modeling of MOSFET drain current. That will have provided you with sufficient knowledge to investigate alternative approaches to modeling and/or to model non-planar bulk-Si transistors. Alternatively, you might propose a project that focuses on device design. In such cases, it will be necessary to demonstrate that the project will go well beyond the material to be presented in class in the latter part of the semester.

Original research projects may be wholly analytic, or may be partly analytic combined with the results of device simulation (e.g., TCAD) or experimentation. Students wishing to do experimental work must provide their own samples.

In a survey paper, state-of-the-art research on a particular topic is reviewed and critiqued. Conclusions should be drawn; for example, if a new device model has been proposed, you should comment on its accuracy relative to published measurement data, its efficiency relative to previous models, and the physical description it presents.

You are encouraged to *propose your own research topic* based on something you've encountered which piqued your interest. However, you may **not** propose a project that you are already working on for thesis credit (ECE 599). If you need assistance identifying a project topic, some possible subjects are listed below. In some cases, I provide a reference to get you started on the topic, but your literature search will need to span far wider than a single paper.

- RF MOSFET model. It is recommended to focus on just one or two aspects of RF modeling. (1) Continuity of derivatives and model symmetry (refer to C. McAndrew, "Validation of MOSFET model source-drain symmetry," *IEEE Trans. Elec. Dev.*, Sept. 2006). (2) Impedance model for the transistor body (an introductory work is by W. Liu et al., "RF MOSFET modeling accounting for distributed substrate and channel resistances with an emphasis on the BSIM3v3 model"). (3) Non-quasi-static transient model (refer to Chapter 9 in the on-reserve textbook by Tsvidis). (4) Noise models.
- Source injection limited drain current models. One such model is the virtual source (VS) model (see A. Khakifirooz et al., *IEEE Trans. on Electron Devices*, Aug. 2009). The VS model extends smoothly from subthreshold to the on-state and cleverly accounts for the fact that the V_T definition changes as the device moves from weak to strong inversion. A more rigorous analysis of the underlying physics can be found in papers from Prof. Mark Lundstrom's group; the earliest of those works is M.

Lundstrom, “Elementary scattering theory of the Si MOSFET,” *IEEE Elec. Dev. Lett.*, Jul. 1997. The above referenced works were inspired by an earlier, very influential paper: K. Natori, “Ballistic metal-oxide-semiconductor field effect transistor,” *J. Appl. Phys.*, Oct. 1994. (A complete understanding of the paper by Natori requires an advanced background in solid-state physics, e.g. ECE 535.) None of the above referenced works present models that are supported by foundries or used for circuit design; their simplicity prevents them from modeling many real-world non-idealities and geometrical effects. However, some of the concepts developed in those models have made their way into the more practical models.

- BSIM-Bulk (also known as BSIM6) (refer to Y. Chauhan et al., “BSIM6: Analog and RF compact model for bulk MOSFET,” *IEEE Trans. Elec. Dev.*, Feb. 2014 and the BSIM6 user’s guide). BSIM-Bulk adopts a charge-based modeling approach to address the RF simulation deficiencies of BSIM3 and BSIM4 and it also incorporates a source injection limit on I_{DSAT} . Students with IC design experience (e.g., from class or summer internship) will almost certainly have used BSIM4 (or even BSIM3) models for circuit simulation—foundries provide BSIM6 models only for very advanced process nodes.
- Surface-potential based model, e.g. PSP and/or HiSIM.
- I-V model for non-planar devices, e.g., FinFET or GAA/nanosheet/ribbon transistor (refer to section 7.2 in the textbook). A well-established model is BSIM-CMG.
- Survey paper on workfunction engineering for metal gate transistors
- Gate current modeling
- Statistical models that account for process variations
- 2D electrostatic analysis for modeling of short channel effect, channel length modulation, etc.
- MOS transistors for high-voltage applications, e.g., lateral drain-extended NMOS. One might attempt to optimize the device doping profile (independent research project) or evaluate the I-V models being used for these devices (survey paper)
- Model parameter extraction techniques (parameters such as L_{eff} and v_{sat})
- I-V model for UTB (ultra-thin body) SOI devices. A well-established model is BSIM-IMG.

Tools

TCAD and HSPICE are available on EWS machines; notify the instructor if you have trouble accessing those tools. Note that you are not required to make use of any such tools.

You may avail yourselves of the Writers’ Workshop for assistance in organizing your project report (see <https://writersworkshop.illinois.edu/locations/> for locations and hours). Grammar checks can be performed by your classmates (i.e., swap papers with a fellow classmate and correct each other’s errors).